Demand Imputation Model Experimental Walkthrough

Uncertainty-Aware Demand and Inventory Planning Tool

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Today's Agenda

Introduction: The Feature & the Task, The Models

Experiment: Data Split, Controlled Conditions, Metrics

Edge Cases and Statistics: Storyboard B Slice, Calibration, Error Analysis

Consequent Actions: Decision Rules, Fallback, Technical Debt

The Feature and the Task: Demand Imputation

- **Input:** hourly sales history for each store-product pair
- **Output:** hourly demand during stockouts, with a 95% range
- **Acts on it:** the demand planner, for sales planning
- **Task:** time-series imputation (de-censoring)

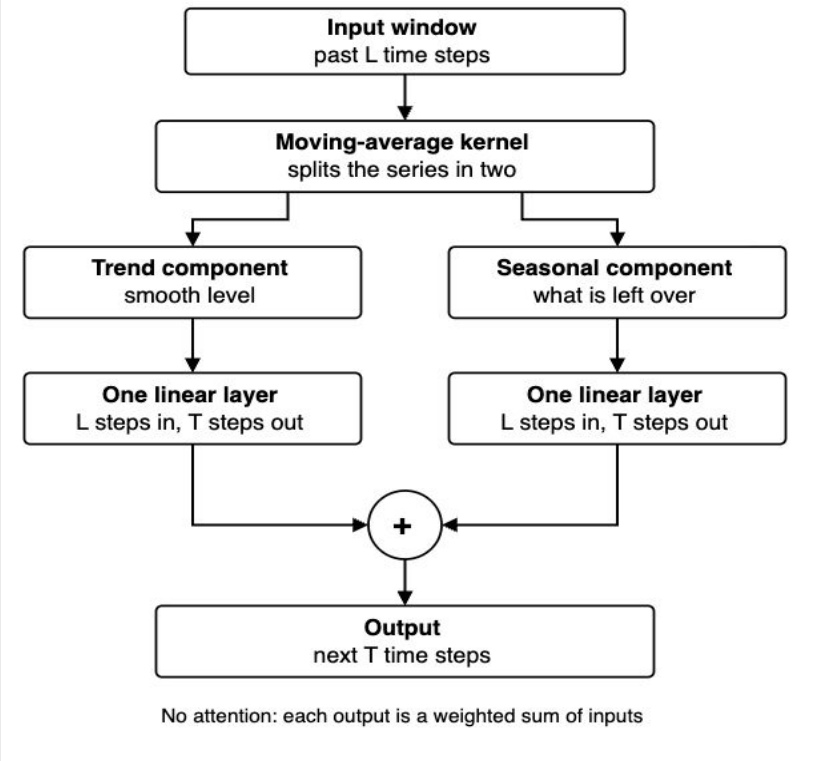


Reconstructed demand: $d = y \odot s + \hat{d} \odot (1 - s)$

Imputation model: $\hat{d} = f_\theta(y \odot s,\ s,\ \text{covariates})$

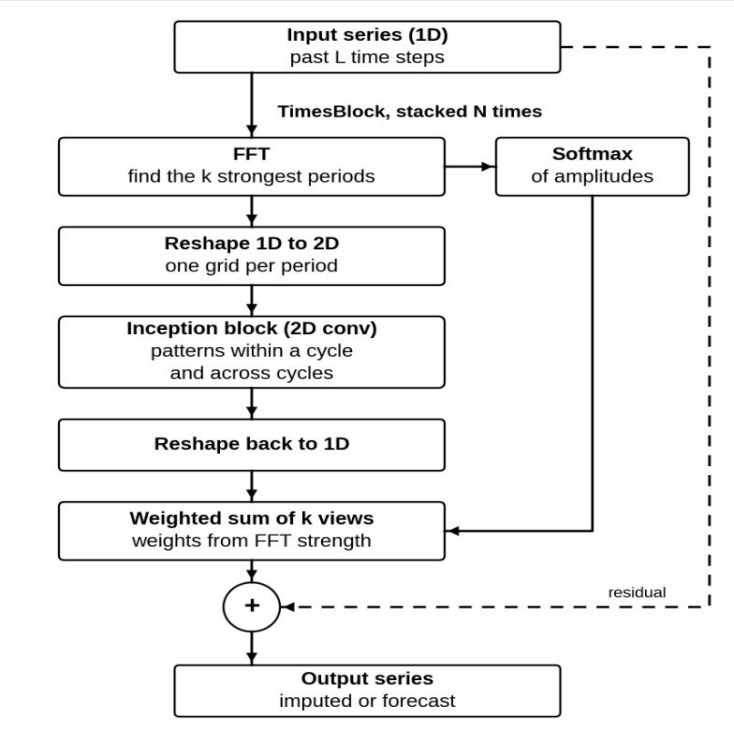
Stock indicator: $s_{j,k,t} = 1$ if item j at store k is in stock at hour t, else 0

DLinear Architecture



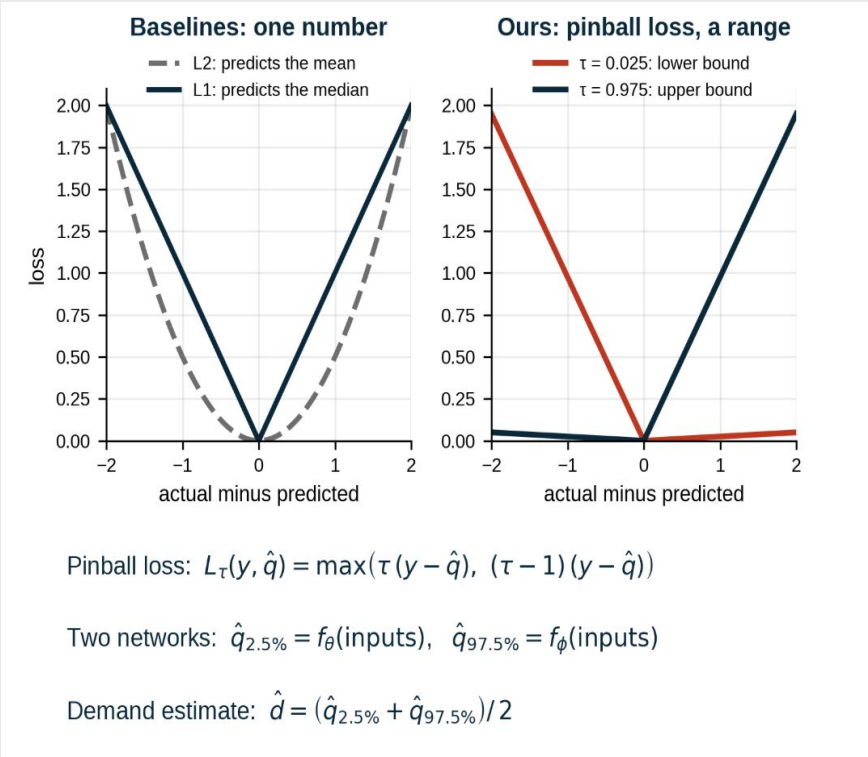
- Splits the input into a trend part and a seasonal part
- One linear layer per part maps L past steps to T future steps; the two are added
- No attention, so it is cheap and its weights are easy to read

TimesNet Architecture



- FFT finds the strongest repeating cycles, such as daily and weekly
- Each cycle is folded into a 2D grid so convolutions see patterns within and across cycles
- Best imputation results in the dataset paper, but heavier and harder to interpret

Main Difference From Baseline: Loss Functions



Pinball loss: $L_\tau(y, \hat{q}) = \max(\tau\,(y - \hat{q}),\ (\tau - 1)\,(y - \hat{q}))$

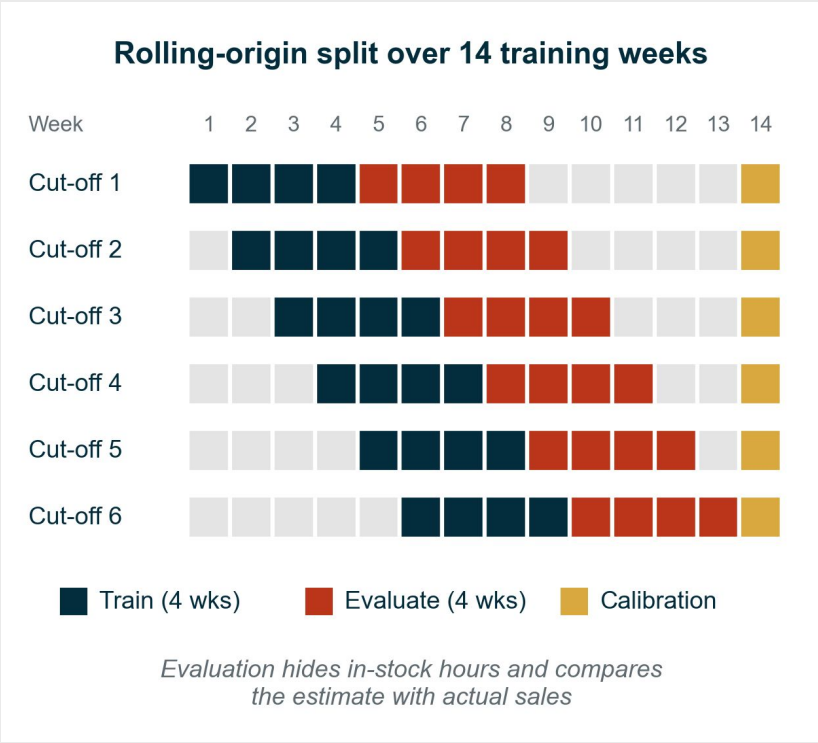
Two networks: $\hat{q}_{2.5\%} = f_\theta(\text{inputs}),\quad \hat{q}_{97.5\%} = f_\phi(\text{inputs})$

Demand estimate: $\hat{d} = (\hat{q}_{2.5\%} + \hat{q}_{97.5\%})/2$

- DLinear and TimesNet use L2 (mean) or L1 (median) loss; TFT uses quantile loss
- Our implementations use 2.5th and 97.5th percentile quantile (pinball) loss for all models
- May need two networks per model, one per bound; the demand estimate is their average

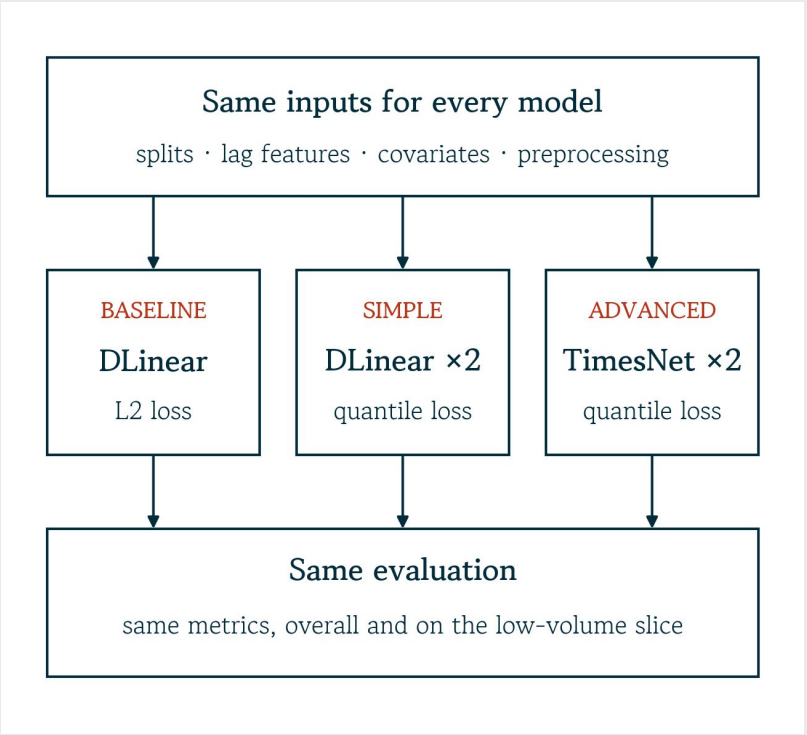
The Data Split: 4 Weeks Training, 4 Weeks Evaluation

- **Rolling origin:** 6 cut-offs across the 14 training weeks
- **Stockouts excluded:** their zero sales are not true demand
- **Calibration:** last week held out for conformal prediction
- **Leakage prevented:** a random split would train on future sales



Controlled Conditions

- **Equal tuning:** dataset-paper settings, equal budget
- **Seeds:** at least 3 per model, reported as mean ± SD
- **Test once:** evaluation weeks are never used for tuning
- **Compute:** same Explorer cluster, training time logged



Metrics and Targets

Metric	What it measures	Formula	Target
WAPE	Size of the error	$100\% \cdot \frac{\Sigma_t \lvert A_t - F_t \rvert}{\Sigma_t A_t}$	Within 1 SD of baseline (29.99%)
WPE	Under- or over-estimation	$100\% \cdot \frac{\Sigma_t (A_t - F_t)}{\Sigma_t A_t}$	Comparable to baseline (−1.28%)
PICP	Coverage of the 95% range	$\frac{1}{N} \Sigma_i 1[\hat{q}_i^{2.5\%} \leq y_i \leq \hat{q}_i^{97.5\%}]$	Mean above 80%
MPIW	Width of the 95% range	$\frac{1}{N} \Sigma_i (\hat{q}_i^{97.5\%} - \hat{q}_i^{2.5\%})$	No fixed target; lower is better
MASE	Error vs. a naive forecast	$\frac{\Sigma_j \lvert A_j - F_j \rvert / h}{\Sigma_t \lvert A_t - A_{t-1} \rvert / (n-1)}$	Client request; for information

Storyboard B: Low-Volume Test Slice

Failure from our Storyboard



Low-Volume Products

- Mean scaled demand below 1
- Frequent zeros complicate recovery
- Evaluate errors separately per model

Datasheet Gap

- Stockout-label verification is unclear
- Zero sales may hide unmet demand

Calibration: Conformalized Quantile Regression

- **Generate**

  By construction, the 2.5th and 97.5th quantiles define a nominal 95% prediction interval.

- **Calibrate**

  Apply conformal prediction using validation data.

- **Evaluate**

  Measure PICP and MPIW on held-out sales, both overall and for low-volume products.

- **Contract criteria**

  Mean PICP > 80%(lowest MPIW preferred)

Error Analysis: Overall and Low-Volume Products

Examine the 10 largest imputation errors per model, overall and separately among low-volume products.

- **Demand Estimation Errors**

  How close was estimated demand to known, masked sales?

- **Low Volume Product Errors**

  Do products with mean scaled demand below 1 have larger estimation errors?

- **Uncertainty Errors**

  Did the 95% prediction interval contain the actual sales?

- **Connection to StoryBoard B**

  Investigate whether suspicious stockout records contributed to underestimated demand.

Decision Rule: Choose the Model with the Sharpest Uncertainty Estimates

- If all models meet prerequisites for performance, choose the model with the smallest MPIW
- Sharp intervals mean more precise information to make decisions with

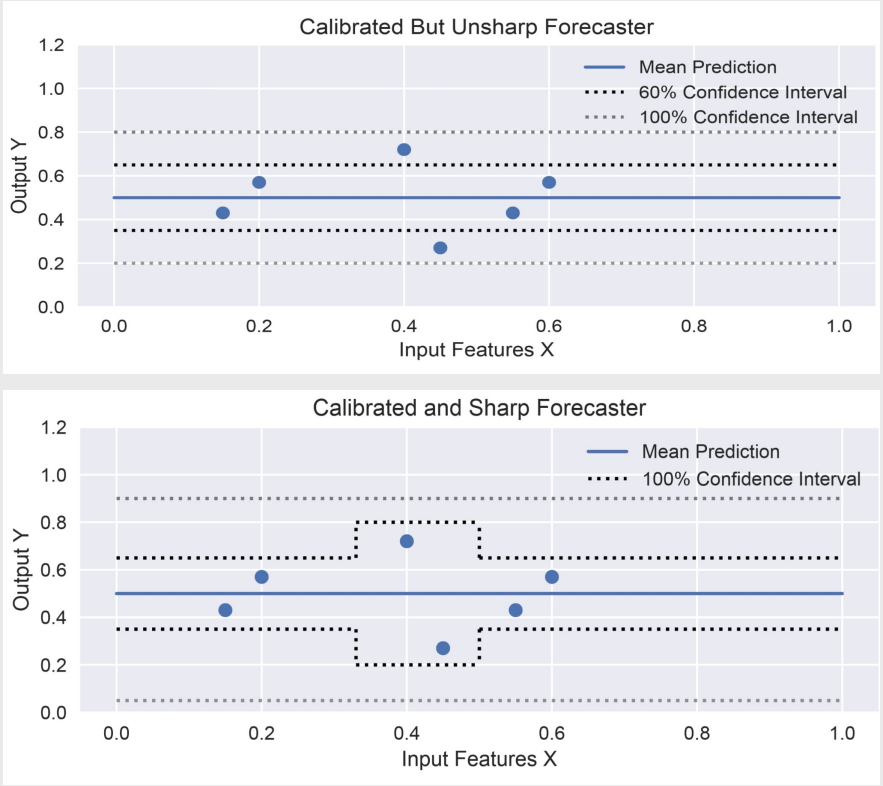
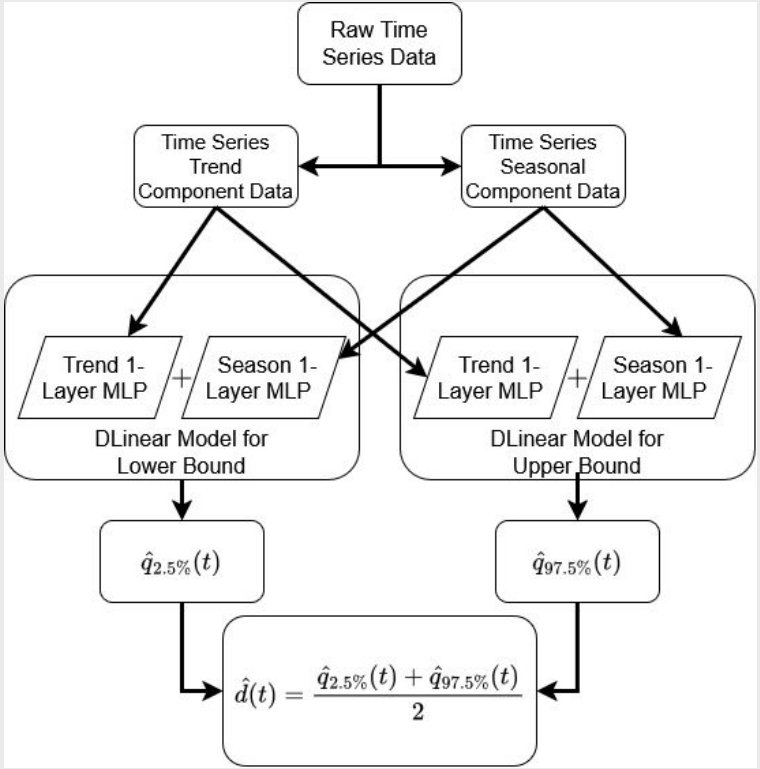


Figure: Accurate Uncertainties for Deep Learning Using Calibrated Regression, Volodymyr Kuleshov, Nathan Fenner, Stefano Ermon, ICML 2018 (link)

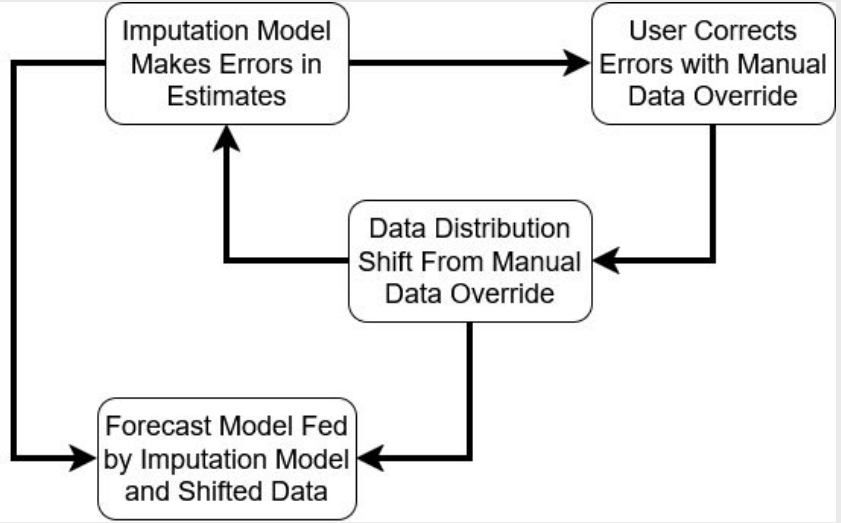
Fallback: Choose the Cheapest and Most Interpretable Quantile Regression Model



- Deployment still requires uncertainty estimate, ruling out baseline mean estimator
- DLinear architecture is more interpretable and cheaper than TimesNet
- Accuracy and uncertainty calibration metrics will be reported on web interface

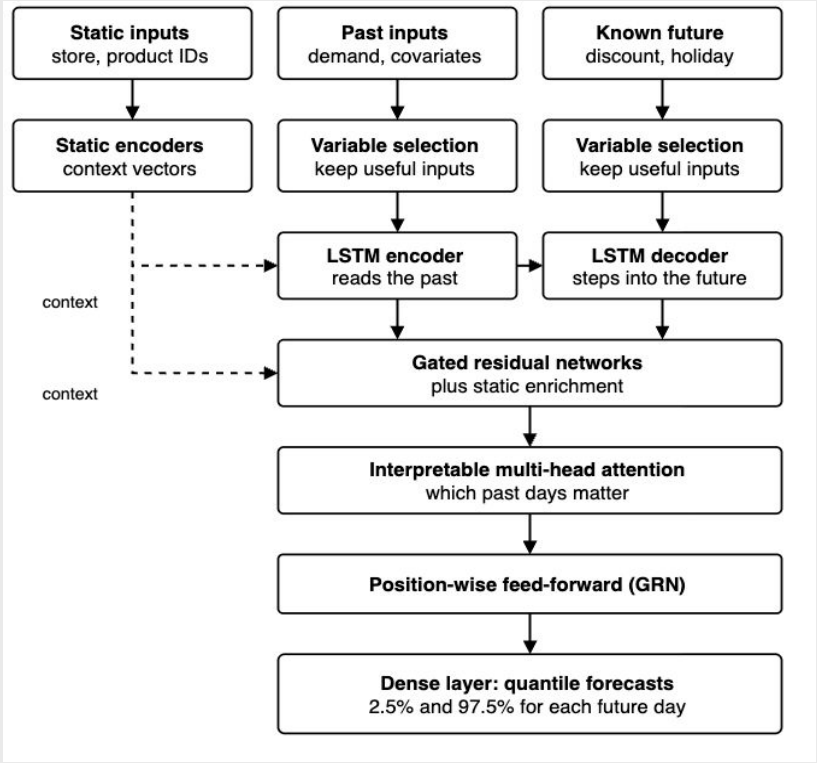
Technical Debt: Unstable Data Dependencies and Hidden Feedback Loops from Manual Overrides

- Data feedback loop exists between manual override and imputation model
- Leads to unstable data dependencies for both imputation and forecasting models
- Models and data must be versioned in deployment to mitigate this



Back-up Slides

TFT Architecture



- Treats static, past, and known future inputs separately and selects the useful ones
- LSTM encoder and decoder plus attention capture short and long patterns
- Outputs quantiles directly; best forecaster in the dataset paper

Field	Entry
Feature Description	**Input**: Hourly aggregate store-product combination sales history. **Output**: Hourly imputed demand for store-product combinations during stockout periods. **Acts On It**: The demand planner, who uses historical data for sales planning.
Task Type	Time series imputation/de-censoring
Architectures Compared	**Baseline**: DLinear architecture trained using the loss and exact architecture from the dataset paper repository (not quantile). **Simple**: Two DLinear models (1-layer feedforward MLP for trend and seasonality) trained to estimate the upper and lower bounds of the 95% confidence interval; chosen for comparability and interpretability. **Advanced**: Two TimesNet models (Fourier-space convolutional networks) trained to estimate the upper and lower bounds of the 95% confidence interval; chosen for comparability and baseline performance.

Split Design	Equal-duration rolling-origin split by time, excluding stockouts: train on the 4 weeks of training-split data prior to a cut-off and evaluate on the 4 weeks after the cut-off, for 6 random cut-off periods within the feasible period (the training data is 14 weeks long). Stockout (0-valued) data are excluded from training and metric calculation because they are not ground-truth demand. A true random split could leak data. The last week of sales is held out as the conformal calibration dataset.
Controlled Conditions	Same lag features, covariates, and cut-offs for all evaluated architectures. Architecture hyperparameters will be kept as similar to those in the dataset paper as possible, so minimal tuning is anticipated. At least three random seeds will be used.
Metrics and Targets	Weighted Absolute Percentage Error (WAPE, also known as MAPE) within 1 standard deviation of the reported baseline (29.99%) as a measure of error magnitude across splits; Weighted Percentage Error (WPE) as a measure of error direction, comparable to the reported baseline (-1.28%); Prediction Interval Coverage Percent (PICP) above 80%; Mean Prediction Interval Width (MPIW; for information only); and the client-requested Mean Absolute Scaled Error (MASE; for information only).

Storyboard B Slice	Product-store combinations with mean scaled demand less than 1, as defined in the dataset authors' GitHub repository.
Calibration	By construction, and by conformalizing the quantile regression models, the models will approach a coverage probability of 95%. This will be evaluated using PICP and assessed with respect to MPIW.
Error Analysis	The ten largest mis-estimations by magnitude overall, as well as those in the low-sales category.
Decision Rule	The model that achieves the prerequisite performance metrics (WAPE, PICP) with the lowest MPIW, with WPE and MASE as tie-breakers.
Fallback	If no architecture qualifies, the simple architecture (conformalized DLinear) will be taken as the reference standard and used in deployment, with explicit mention of its performance, due to low computational cost and inherent interpretability.
Technical Debt	Planners and stockers are anticipated to manually override demand estimates they find unsupported. This will likely create a hidden feedback loop that would surface during future re-training after the proof of concept is deployed.

What Is Conformal Prediction?

- An approach to taking any statistical model with a heuristic notion of uncertainty and making it rigorous

- Using a holdout dataset for calibration of uncertainty estimates, calibrated prediction sets are guaranteed to satisfy

$$\mathbb{P}[Y_{test} \in \mathcal{C}(X_{test})] \geq 1 - \alpha$$

Conformal Prediction Steps

- Identify a heuristic notion of uncertainty output by a statistical model
- Define a score function that measures agreement between model input, model output, and actual output label
- Compute scores for calibration data of size n as well as the appropriate quantile of those scores at significance level alpha
- Use that quantile to form prediction sets

$$\hat{q} = \frac{\lceil (n+1)(1-\alpha) \rceil}{n}$$

$$\mathcal{C}(X_{\text{test}}) = \{y : s(X_{\text{test}}, y) \leq \hat{q}\}$$

Conformalized Quantile Regression

$$\alpha = 0.05 = 5\%$$

$$\begin{aligned}\text{Score}(x, y) &= \min\{y - \hat{q}_{\frac{\alpha}{2}}(x), \hat{q}_{1-\frac{\alpha}{2}}(x) - y\} \\ &= \min\{y - \hat{q}_{2.5\%}(x), \hat{q}_{97.5\%}(x) - y\}\end{aligned}$$

$$q_{\text{score}} = \text{Quantile}\left(\{\text{Score}(X_j, Y_j) | j \in [n]\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$$

$$\mathcal{C}(x) = \{\hat{q}_{2.5\%}(x) - q_{\text{score}}, \quad \hat{q}_{97.5\%}(x) + q_{\text{score}}\}$$

- Heuristic Uncertainty: 95% confidence interval estimate
- Score: distance between data and closest interval bound
- Calibration data: last week of sales with non-zero ground truth

$$\begin{aligned}\mathbb{P}[Y_{test} \in \{\hat{q}_{2.5\%}(X_{test}) - q_{\text{score}}, \\ \hat{q}_{97.5\%}(X_{test}) + q_{\text{score}}\}] \geq 1 - \alpha = 95\%\end{aligned}$$